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Geophysical evidence for widespread reversely magnetised pyroclastics in the western Taupo Volcanic Zone (New Zealand)

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Abstract Low-altitude aeromagnetic data show that negative residual anomalies are widespread over the western Taupo Volcanic Zone, New Zealand. Paleomagnetic study of eight rhyolitic ignimbrite units and two lava flows which are exposed in this area, together with new K-Ar dates of four of the ignimbrite units, indicate that the two lava units and seven of the ignimbrite units were erupted during the Matuyama geomagnetic epoch (>0.73 Ma B.P.) and suggest that rhyolitic volcanism in the western Taupo Volcanic Zone began as early as 1.6 Ma B.P. These results provide the basis for an interpretation of our aeromagnetic data which confirms the hypothesis that the magnetic anomalies observed in the western Taupo Volcanic Zone are caused by widespread, thick, reversely magnetised pyroclastic and lava flows. Magnetic modelling also allows thickness estimates of the younger, normally magnetised cover rocks which reach a maximum thickness of the order of 0.5 km in the Mangakino area. The magnetic structure of these volcanic rocks defines approximately the lateral extent of the Mangakino Volcanic Centre.

Keywords Western Taupo Volcanic Zone; K-Ar dates; pyroclastic flows; paleomagnetism; stratigraphy; low-level aeromagnetic anomalies; residual magnetic anomalies

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INTRODUCTION

The Taupo Volcanic Zone (TVZ) is a region of Quaternary volcanic and geothermal activity in the central North Island, New Zealand. Since volcanism commenced in this region, more than 16 000 km^3 of pyroclastics and lavas (dominantly of rhyolitic composition) have been erupted from the central segment of TVZ (Healy 1964). Paleomagnetic studies by Cox (1971) suggested that only normally magnetised (<0.73 Ma old) volcanic rocks occur at the surface of the TVZ. Prior to the present study, which began in 1986, it was believed that all rhyolites outcropping within the TVZ were erupted during the last 1.1 Ma (e.g., Wilson et al. 1984) and that most of them were younger than 0.73 Ma (e.g., Rogan 1982).

Between 1950 and 1952, the Department of Scientific and Industrial Research (DSIR) conducted high-altitude regional aeromagnetic surveys (1500 m a.s.l.; about 3.6 km line spacing) over the TVZ; the results were presented by Gerard & Lawrie (1955) as contour maps of the observed total magnetic force. Residual magnetic anomalies (ΔF) were computed much later by Whiteford (1976) and Hunt & Whiteford (1979a, b) using the regional geomagnetic field of Reilly & Burrows (1973) and then by Rogan (1980, 1982) who used the normal geomagnetic field of Reilly et al. (1978). The ΔF data of the two maps show up with nonzero values over nonmagnetic Mesozoic greywacke basement, reflecting the effect of a regional field. Although the maps contain no detailed short wavelength anomalies, which are important for interpreting shallow structures, they indicate that some negative ΔF anomalies are associated with the western TVZ.

LOW-ALTITUDE AEROMAGNETIC SURVEYS OVER THE WESTERN TVZ

In May 1986 and March 1988 we conducted low-altitude (760 m a.s.l.; 100–400 m ground clearance) aeromagnetic surveys with about 1 km spacing between flight lines to map in more detail the magnetic anomaly pattern over the western TVZ. Data from these surveys were reduced using the global International Geomagnetic Reference Field (Malin & Barraclough 1981). This global geomagnetic field reproduces in detail secular variations giving the same anomaly values over the same area for data collected in the TVZ between 1950 and 1988. However, residual ΔF anomalies of the 1986/88 survey still attained nonzero values over the nonmagnetic basement. By upward continuation of the two data sets observed at 1.5 and 0.76 km height, a long wavelength regional field could be defined for the TVZ (Soengkono 1990). Subtraction of this field produced zero value anomalies over the nonmagnetic basement and provides a representative zero level for all reduced magnetic anomalies over the TVZ.

The reduced ΔF anomalies of the 1986 and 1988 surveys are shown in Fig. 1. This map shows several significant negative anomalies in the western TVZ which are not shown in the old map based on the data from Gerard & Lawrie

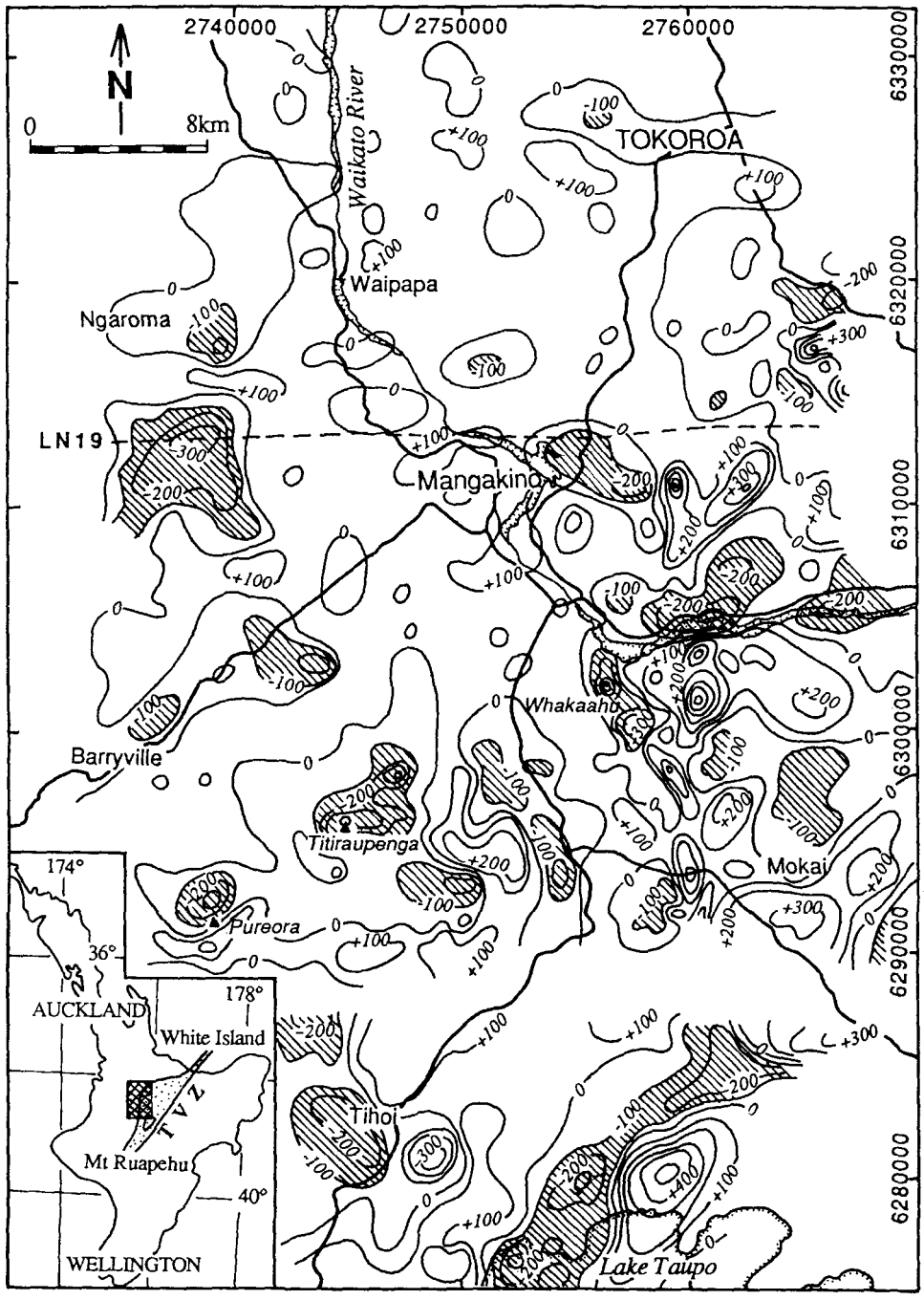


Fig. 1 Residual total force magnetic anomalies over the western TVZ. Areas with significant negative (≤−100 nT) residual anomalies are shown (hatched). Grid coordinates along the edges of the map are in terms of the N.Z. Map Grid (m).

(1955), that is, the negative anomalies northeast of Barryville, near Pureora, between Titiraupenga and Mokai, east of Tihoi, north of Lake Taupo, east of Mangakino, and northeast of Whakaahu.

Although some of the anomalies in Fig. 1 clearly reflect topographic effects, namely those associated with the old volcanoes of Pureora and Titiraupenga, most of the others cannot be explained in terms of exposed volcanic domes. Several models were put forward to explain the origin of the negative anomalies:

(1) The anomalies are caused by isolated, mostly concealed, volcanic deposits which are reversely magnetised; the area between is covered by a thick layer of normally magnetised pyroclastic and lava flows.

(2) The western TVZ is covered by thick, normally magnetised pyroclastic and lava flows, and most negative anomalies in Fig. 1, apart from the few caused by topographic effects, reflect local demagnetisation of ancient geothermal reservoirs. Negative magnetic anomalies have been observed over many active geothermal fields in the central and eastern TVZ which are surrounded by unaltered, normally magnetised volcanic rocks (Hochstein & Hunt 1970; Ignacio 1985; Soengkono 1985; Salt 1986; van Dijck 1988).

(3) The negative anomalies are caused by a thick deposit of reversely magnetised rocks which, in turn, is overlain in some areas by younger, normally magnetised rocks of variable thickness.

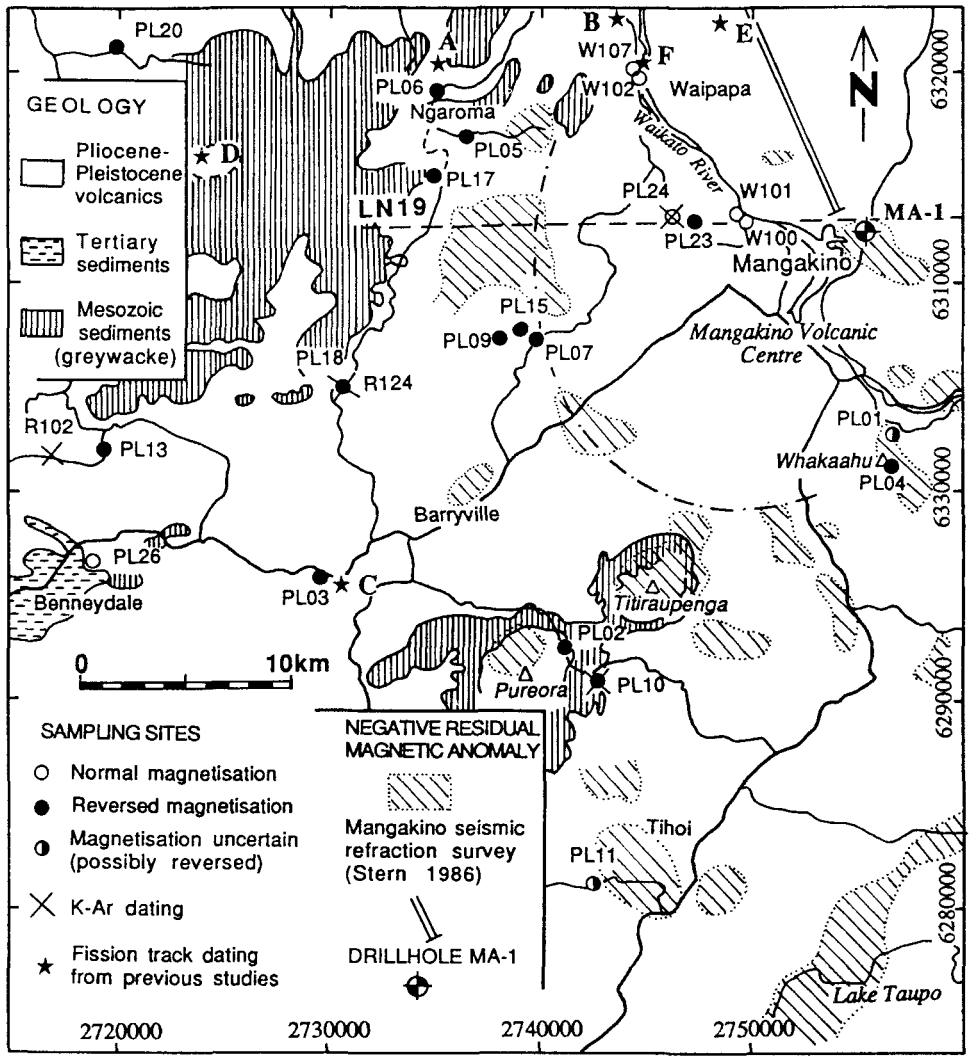


Fig. 2 Map of the study area showing locality of sampling sites. A simplified geological map is superimposed on the significant negative residual magnetic anomaly from Fig. 1. The boundary of the Mangakino Volcanic Centre (Wilson et al. 1984) is also shown. Grid co-ordinates along the edges of the map are in terms of the N.Z. Map Grid (m).

Apart from a single 0.6 km deep well near Mangakino (MA–1 in Fig. 2), there are no deep drillholes in the western TVZ area shown in Fig. 2. Therefore, to check which of these explanations applies, surface samples were collected from exposed solid pyroclastic and lava units covering mainly flows which had been studied in detail by Wilson (1986a). The aim of the sampling program was to establish a gross magnetic stratigraphy of pyroclastic flows in the study area.

Sample sites are shown in Fig. 2. The direction and magnitude of the total magnetisation of all samples were determined by measurement of induced (IM) and natural remanent magnetisation (NRM); these data were used later to obtain representative means of the total magnetisation when the aeromagnetic data were interpreted in terms of source bodies. The origin and stability of the NRM were not studied in detail except for a few samples where the NRM magnitude was either small or giving divergent values for the same flow unit. In this case, the direction and magnitude of the thermal remanent magnetisation (TRM) were also determined.

When we found that the paleomagnetic polarity of several flow units could not be correlated with published radiometric ages of similar rocks inside and outside the study area, we started a separate small K-Ar dating project. For this, we sampled most of the flow units in the study area but found that only a few ignimbrites yielded fresh and unaltered mineral separates suitable for K-Ar dating. Unfortunately most sites from which magnetic samples had been taken did not provide suitable unaltered mineral separates. We only sampled flows which had been mapped in detail by Wilson (1986a, b) to avoid errors in stratigraphic correlation. The K-Ar dates of our few samples, however, were sufficient to confirm the magnetic stratigraphy of the whole sequence of pyroclastic flows exposed in the study area.

PALEOMAGNETIC MEASUREMENT AND K-Ar DATING OF VOLCANIC ROCKS EXPOSED IN THE WESTERN TVZ

Rhyolitic deposits and andesitic lavas exposed along the western margin of the TVZ are products of an early volcanic activity in the zone. Their gross stratigraphic relationship was first established by Martin (1961) and Blank (1965) and has been reviewed by Wilson (1986a, b). Wilson identified eight old pyroclastic units in the following order: Ignimbrite A, B, C, Ongatiti Ignimbrite, unit D, Ahuroa Ignimbrite, unit E, and Rocky Hill Ignimbrite. The units were inferred to range in age from 1.1 Ma to 0.5 Ma and to be related to activity from an ancient Mangakino Volcanic Centre (VC). Another group of ignimbrites, identified as Ignimbrite F, G, H, I (Wilson 1986a,

b), and the Whakamaru-group ignimbrites (Wilson et al. 1986), were thought to range in age from c. 0.5 to 0.3 Ma and to have come from the Maroa and Taupo VC. The ages were based mainly on published fission track (FT) dates (Kohn 1973; Rutherford 1976; Murphy & Seward 1981) and on relative stratigraphic position.

Eight ignimbrite units from the western TVZ were sampled for the paleomagnetic study; in addition, lava samples were taken from Mt Pureora and the Whakaahu dome (see Fig. 2). At each site, at least three oriented, 2.5 cm diameter cores were collected using a portable rock drill. Between one and four oriented, 2 cm long specimens were cut from each core. The NRM of all specimens was measured using the Auckland University spinner magnetometer constructed by Robertson (1976); measurement of IM was conducted using a BISON susceptibility meter. A mean NRM direction was computed for each site using the statistical method of Fisher (1953); the results are listed in Table 1. The stable TRM was determined for a few samples by stepwise removal of all secondary components of remanent magnetisation in an AC field, using a two-axis tumbler system shielded by three-layers of mu-metal, and a Natsuhara Giken's SMM–85 spinner magnetometer at Osaka University, Japan. The data in Table 1 are sufficient to assess the paleomagnetic polarity of the rock units.

K-Ar ages were determined on hornblende separates from Ignimbrite F, the Ongatiti and Rocky Hill Ignimbrites, and on a biotite separate from an ignimbrite flow thought to be a member of the Whakamaru-group ignimbrites. Mineral separations were also attempted for Ignimbrites A and C, Ahuroa Ignimbrite, and Ignimbrite I, but failed to produce significant hornblende or biotite separates. N.Z. Map Grid co-ordinates of the locality of dated samples and their K-Ar dates are listed in Table 2a; sample sites are shown in Fig. 2. The K-Ar dating was conducted at the Okayama University of Science (Japan). Analytical procedures used for determination of K and Ar have been reported (Nagao & Itaya 1988; Itaya & Nagao 1986, 1988). The accuracy and reproducibility of K

Table 1 Summarised results of magnetisation and density measurements of ignimbrite and lava samples from the western Taupo Volcanic Zone.

	N.Z. Map Grid co-ordinates		Mean NRM						TRM				Paleo polarity	Mean IM (Ind. Magn.)	
Site no.	East (m)	North (m)	$n1$	Decl (°)	Incl (°)	α_{95} (°)	$\lvert\underset{\sim}{\mathbf{I}}_{rem}\rvert$ (A/m)	σ (A/m)	$n2$	Decl (°)	Incl (°)	Stab.		$\lvert\underset{\sim}{\mathbf{I}}_{ind}\rvert$ (A/m)	σ (A/m)
Whakamaru Ignimbrites:															
W100	2749500	6313300	7	16.3	–66.7	1.65	2.39	0.16	–				N	0.65	0.04
W101	2749100	6313500	13	8.8	–63.4	1.60	1.54	0.26	–				N	0.41	0.03
W102	2744600	6320200	7	15.0	–69.2	2.87	2.66	0.15	–				N	0.72	0.02
W107	2744500	6313500	7	2.0	–71.3	3.27	1.59	0.06	–				N	0.47	0.01
Unnamed ignimbrite:															
PL24	2746300	6313500	26	13.2	–61.6	1.83	2.43	0.39	–				N	0.43	0.01
Ignimbrite I:															
PL23	2747100	6313100	12	175.7	+67.1	4.36	1.96	0.17	1	188	+65	v.s.	R	0.50	0.01
Rocky Hill Ignimbrite:															
PL26	2718900	6296750	10	13.9	–50.7	3.73	0.33	0.26	–				N	0.43	0.05
Ahuroa Ignimbrite:															
PL07	2739500	6307550	14	351.4	+22.9	6.53	0.30	0.04	2	346	+82	m.s.	R	0.48	0.03
PL13	2719050	6302050	18	47.5	+82.6	2.83	1.15	0.16	2	91	+82	s.	R	0.45	0.02
Ongatiti Ignimbrite:															
PL09	2738000	6307550	12	184.3	+70.9	2.80	0.75	0.12	1	200	+64	v.s.	R	0.48	0.04
PL03	2730100	6295900	17	148.3	+61.5	11.38	0.24	0.07	–				R	0.48	0.05
PL11	2742100	6281200	11	45.8	+60.6	9.02	0.55	0.66	–				R(?)	0.59	0.03
PL18	2729900	6304550	6	171.5	+81.0	12.00	0.24	0.14	–				R	0.40	0.03
PL20	2719800	6321750	10	183.6	+60.1	11.27	1.07	0.13	–				R	0.51	0.03
Ignimbrite C:															
PL15	2738700	6307800	9	177.3	+60.2	8.41	1.44	0.16	–				R	0.42	0.03
Ignimbrite A:															
PL05	2736350	6317300	10	178.3	+55.6	4.08	1.24	0.95	1	194	+57	v.s.	R	0.42	0.06
PL17	2735200	6215000	9	185.9	+70.6	3.55	0.54	0.09	–				R	0.39	0.04
PL06	2734600	6219100	3	193.3	+55.9	8.49	1.57	0.23	–				R	0.42	0.01
Ignimbrite F:															
PL10	2742500	6291000	16	175.1	+66.3	3.74	1.54	0.14	–				R	0.54	0.06
Whakaahu rhyolite:															
PL04	2756400	6301600	6	140.0	+86.6	6.63	0.84	0.13	1	122	+78	s.	R	0.32	0.05
PL01	2756600	6303000	7	256.6	+51.8	23.67	0.61	1.22	1			u.	R(?)	0.24	0.06
Titiraupenga Andesite:															
PL02	2741000	6292400	10	132.8	+41.1	7.22	1.24	0.16	2	168	+60	s.	R	0.74	0.02

$n1$ is the number of specimens used for NRM determination; mean NRM directions were computed using the statistical method of a unit sphere set out by Fisher (1953); α_{95} is the radius of circle of 95% confidence. Error bounds (σ) presented for $\lvert\underset{\sim}{\mathbf{I}}_{rem}\rvert$ and $\lvert\underset{\sim}{\mathbf{I}}_{ind}\rvert$ are the standard deviation.

The direction of TRM is listed separately; $n2$ denotes the number of specimens subjected to stepwise AC demagnetisation in fields up to 70 mT (700 Oersteds). Stability (Stab.) of inferred TRM is classified as: v.s. = very stable, s. = stable, m.s. = moderately stable, u. = unstable. Symbols for the paleomagnetic polarity are: N = normal polarity, R = reversed polarity.

determination was within 2%, and an average value of duplicate runs was used in age calculation. Ar was analysed using an isotopic dilution method and ^{38}Ar spike; reactive gases were cleaned by a Ti-Zr scrubber. Mass discrimination was checked with atmospheric Ar several times each day.

APPROXIMATE STRATIGRAPHY OF VOLCANIC ROCKS IN THE WESTERN TVZ

When we tried to construct a stratigraphic sequence for pyroclastic flows in the western TVZ, we encountered two main problems:

1. Ages of volcanic rocks had been determined prior to our survey at six sites inside the study area (see Fig. 2 and Table 2b). Five dates were obtained by the FT method using glass shards (Kohn 1973); the age of Ignimbrite A was obtained from FT dating of zircon separates (cited in Rutherford 1976). These measurements were made during the pioneering days of FT dating before it was known that dating of glass shards can contain significant errors. There is good evidence that devitrified glass is common in the Ahuroa Ignimbrite (R. M. Briggs pers. comm.) and it is almost certain that the FT age of this flow (0.65 Ma) given by Kohn contains an error. It is also possible that a flow unit sampled by Kohn (1973) inside the study area was not correctly identified. One of his samples (D in Table 2b) was taken from an outcrop described as "Rocky Hill Ignimbrite"; it has been pointed out by Wilson (1986a) that this sample was probably taken from one of the younger Whakamaru-group ignimbrites. In view of recent developments in the understanding of the kinetics of annealing of fission tracks in apatite (Green et al. 1986), developments in the calibration of FT ages against other geochronological dating systems (Green 1985), and improvements in the error analysis (Green 1981), it appears that FT ages of glass obtained during the pioneering days of FT dating should be considered with caution. Because of the relative instability of glass (compared with apatite), the glass ages are probably minimum estimates and their associated errors may have been underestimated.

To overcome these problems, we discarded all existing FT dates of glass in the study area and constructed a stratigraphic column based only on our own data and stratigraphic field observation.

2. It is likely that different pyroclastic flows in the greater western TVZ have been misidentified. Wilson et al. (1984) pointed out that Marshall Ignimbrites sampled in the Matahana Basin by Murphy & Seward (1981) cannot be correlated with the Marshall Ignimbrites in the Mangakino area which were subdivided by Wilson (1986a) into three units (Ignimbrite I, H and G). We could not obtain any suitable mineral separates for K-Ar dating from any of these units; oriented cores could only be obtained for the youngest unit (Ignimbrite I), and the magnetic studies showed clearly that this unit is reversely magnetised (i.e. age >0.73 Ma). In a drillhole near Tokoroa, Ignimbrite I is underlain by Rocky Hill Ignimbrite (Houghton et al. 1987). To the north of Tokoroa, Marshall Ignimbrites were found to be normally magnetised (Cox 1971); FT dates on zircon of 0.52 Ma for these flows have been reported in the Matahana Basin, about 20 km to the east of Tokoroa (Murphy & Seward 1981).

Table 2a Results of K-Ar dating.

Rock unit	Mineral dated	Site no.	N.Z. Map Grid co-ordinates East (m)	N.Z. Map Grid co-ordinates North (m)	K (wt%)	Rad. Ar^{40} (10^{-6} mm^3 STP/g)	K-Ar age (Ma)	Non rad. Ar (%)
Whakamaru-ignimbrite (?)	bi	PL24	2746300	6313500	4.68 ± 0.09	111.0 ± 2.4	0.61 ± 0.13	93.1
Rocky Hill Ignimbrite	ho	R102	2716300	6302000	0.90 ± 0.03	37.1 ± 0.29	1.06 ± 0.09	82.6
Ongatiti Ignimbrite	ho	R124	2729900	6304550	0.72 ± 0.02	35.0 ± 0.24	1.25 ± 0.09	79.9
Ignimbrite F	ho	PL10	2742500	6291000	0.49 ± 0.02	30.4 ± 0.37	1.60 ± 0.21	88.3

In the second column "bi" refers to biotite, "ho" to hornblende.

Table 2b Results of fission track dating from previous studies in the study area.

Rock unit	Mineral dated	Site no. (refer to Fig. 2)	N.Z. Map Grid co-ordinates East (m)	N.Z. Map Grid co-ordinates North (m)	Fission-track age (Ma)	Reference
Ignimbrite A	zircon	A	2734760	6320520	1.1*	Rutherford (1976)
Ahuroa Ignimbrite	glass	B	2743240	6322750	0.65 ± 0.09	Kohn (1973)
Rocky Hill Ignimbrite**	glass	C	2730130	6295960	0.31 ± 0.06	Kohn (1973)
Rocky Hill Ignimbrite	glass	D	2723560	6315620	0.33 ± 0.05	Kohn (1973)
Marshall Ignimbrite	glass	E	2748470	6323330	0.63 ± 0.09	Kohn (1973)
Whakamaru-group ignimbrites	glass	F	2744900	6320600	0.33 ± 0.05	Kohn (1973)

*No standard deviation quoted.
**Suggested by Wilson (1986a) to be one of the younger Whakamaru-group ignimbrites.

We therefore did not try to correlate pyroclastic flows sampled and mapped inside the study area (Fig. 2) with apparently similar flow units outside the area; some FT dates from outside the area are listed in italics in Table 3. The dates for the Ongatiti Ignimbrite and Whakamaru-group ignimbrites (except the mean 0.35 Ma quoted by Grindley et al. 1988) are also based on FT measurement of volcanic glass.

By using the magnetic data listed in Table 1, the K-Ar dates on biotite and hornblende separates listed in Table 2a, and the relative stratigraphic position of flow units as observed in the field, we could erect the stratigraphic sequence shown in Table 3. Although we could not obtain suitable samples for magnetic and radiometric studies from some friable pyroclastic flows (namely unit H, G, E, D and Ignimbrite B), the sequence shown in Table 3 is well constrained by our data.

When we compiled the sequence we found a few interesting results which warrant discussion.

1. Our sample PL24 was taken from the basal layer of a welded ignimbrite containing significant black glassy fiammé which was previously described by Wilson (1986b) as one of the Whakamaru-group ignimbrites. However, its age of 0.61 Ma (see Table 2a) is significantly older than that of Whakamaru-group ignimbrites. Our magnetic interpretation model (Fig. 3) shows that this site lies close to the contact of normally and reversely magnetised volcanic rocks. Since we have no reason to doubt the accuracy of our K-Ar dates, which are based on measurement of fresh, unaltered minerals, it appears that sample PL24 belongs to an older, until now unknown, ignimbrite unit which lies beneath the Whakamaru-group ignimbrites.

2. Site R102, from which a sample of Rocky Hill Ignimbrite was collected for K-Ar dating, was unsuitable for drilling of an oriented core. The magnetic polarity of this flow unit was obtained from a core at site PL26. Both sites have been mapped as Rocky Hill Ignimbrite by Wilson (1986a) who also showed us both sites in the field. Although the K-Ar age of 1.06 ± 0.09 Ma of this flow appears to be older than that indicated by its paleomagnetic normal polarity, we can reconcile both results by assuming that this flow was deposited during the Jaramillo geomagnetic epoch (i.e. between 0.90 and 0.97 Ma ago; Mankinen & Dalrymple 1979). Since the K-Ar age of the hornblende minerals dates crystallisation (Cliff 1985), whereas magnetic polarity relates to the date of extrusion, we believe that the small difference between the two dates is not significant.

INTERPRETATION OF THE RESIDUAL AEROMAGNETIC ANOMALIES

The results of the paleomagnetic and dating study show that the first hypothesis, namely, that the negative magnetic anomalies in the western TVZ are caused by isolated, reversely magnetised deposits, can be rejected. Since no areas with coherent alteration were found, the second explanation model could also be rejected. The field study supports the third

Table 3 Tentative stratigraphic sequence of the western Taupo Volcanic Zone.

Name of unit	Fission-track age (Ma)	K-Ar age (Ma)	Observed paleomagnetic polarity	Source area
Whakamaru-group ignimbrites	0.33 ± 0.05[2] *0.23*[5] *0.35 ± 0.16*[6]		normal[8]	Maroa or Taupo VC
Unnamed ignimbrite (oldest member of the Whakamaru-group ignimbrites ?)		0.61 ± 0.13[8]	normal[8]	?
Marshall Ignimbrites (Matahana Basin, Tokoroa, Waipapa)	*0.52 ± 0.14*[4] 0.63 ± 0.09[2]	n.a.	*normal*[7]	?
0.73 Ma — Brunhes (N) and Matuyama (R) boundary				
Marshall Ignimbrites (Mangakino area):				
Unit I	n.a.	n.a.	reversed[8]	?
Unit H	n.a.	n.a.	n.a.	?
Unit G	n.a.	n.a.	n.a.	?
Rocky Hill Ignimbrite	0.33 ± 0.05[2]	1.06 ± 0.09[8] (Jaramillo event)	normal[8]	Mangakino VC
Unit E	n.a.	n.a.	n.a.	?
Ahuroa Ignimbrite	0.65 ± 0.09[2]	n.a.	reversed[8]	Mangakino VC
Unit D	n.a.	n.a.	n.a.	?
Ongatiti Ignimbrite	*0.75 ± 0.08*[2]	1.25 ± 0.09[8]	reversed[8]	Mangakino VC
Ignimbrite C	n.a.	n.a.	reversed[8]	Mangakino VC
Ignimbrite B	n.a.	n.a.	n.a.	Mangakino VC
Ignimbrite A	1.1[3]	n.a.	reversed[8]	Mangakino VC
Ignimbrite F	n.a.	1.60 ± 0.09[8]	reversed[8]	East of Titiraupenga (?)
Titiraupenga Andesite	n.a.	1.9[1]	reversed[8]	local

References: [1]Stipp (1968); [2]Kohn (1973); [3]Seward quoted in Rutherford (1976); [4]Murphy & Seward (1981); [5]Kohn quoted in Wilson et al. (1984); [6]Grindley et al. (1988); [7]Cox (1969); [8]this study; n.a. denotes that data are not available.

Source areas are from Wilson (1986b). Figures and text in *italics* give fission track ages and polarity from sites outside the western TVZ shown in Fig. 2.

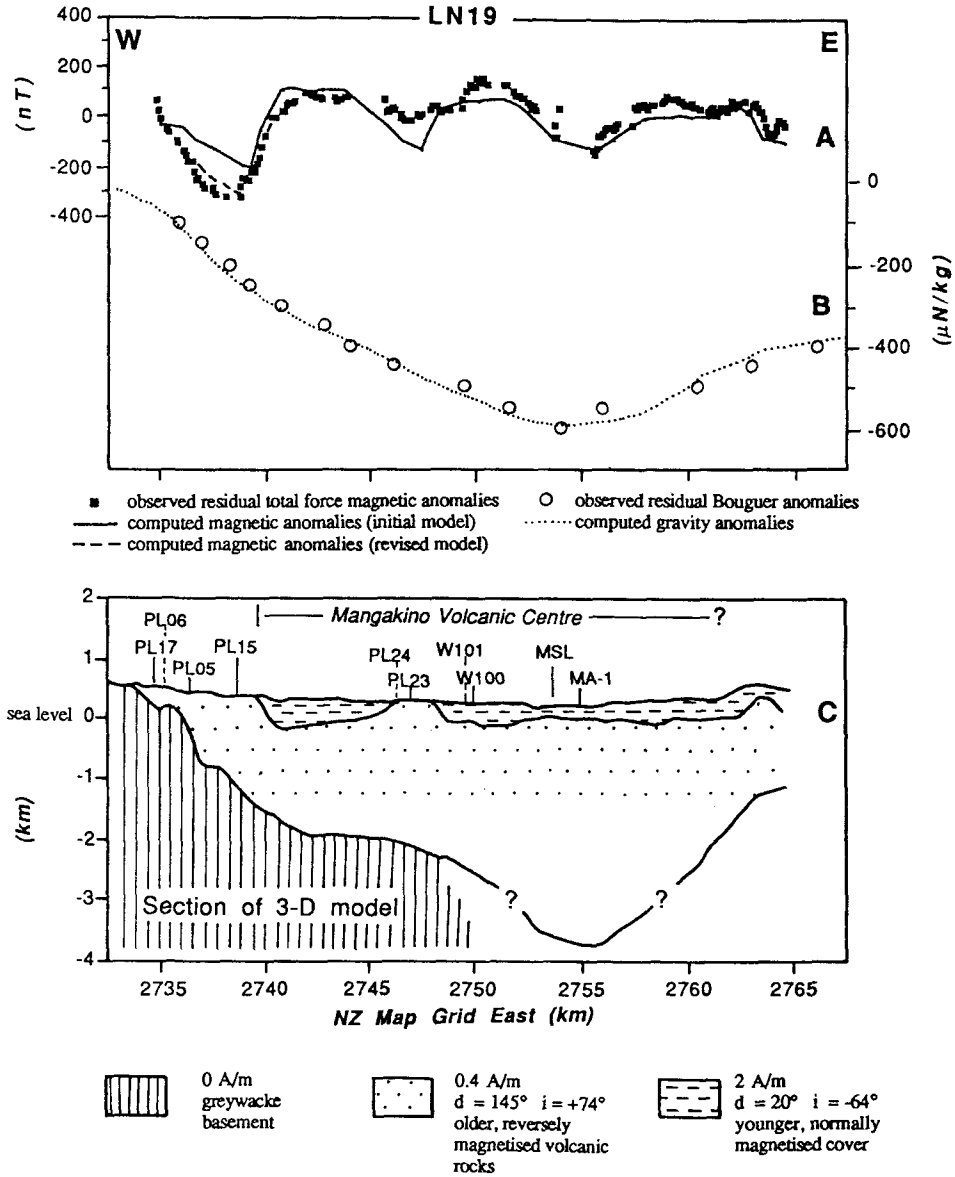


Fig. 3 Observed and computed (3-D model) magnetic (A) and gravity (B) profiles over the western TVZ (upper part of figure). Localities (projected) of some paleomagnetic sampling sites (Fig. 2) are shown in the model section (C) in the lower part. MSL = southern end of Mangakino seismic refraction line (Stern 1986); MA-1 = Mangakino drillhole. The dashed line on the left-hand side of the magnetic profile (A) denotes magnetic effects of the model computed using a magnetisation of 0.8 A/m with d = +182° and i = +56° (i.e. the average total magnetisation of Ignimbrite A and C sampled at sites PL17, PL06, PL05, and PL15) for reversely magnetised rocks in the middle layer. Density contrast of volcanic rocks with respect to greywacke basement is –500 kg/m^3. The location of the profile is shown in Fig. 1 and 2.

hypothesis, for which widespread, reversely magnetised volcanic rocks were assumed. The final proof for the third model arises from a quantitative interpretation of the observed residual aeromagnetic anomalies.

For the quantitative interpretation, rocks from the western TVZ area were subdivided on the basis of their magnetic properties into: (1) the substratum of nonmagnetic Mesozoic sediment basement (greywacke); (2) a middle group of reversely magnetised rocks (Rocky Hill Ignimbrite being the only exception to this); and (3) a normally magnetised upper layer. Results of remanent and induced magnetisation measurements of the welded part of ignimbrite units listed in Table 1 provide estimates of the representative total magnetisation for the normally (upper) and reversely (middle) magnetised layers. The mean value of total magnetisation of the upper layer (vector average of data in Table 1) is 2.6 A/m; this value, however, is biased towards the high values of the welded Whakamaru-group ignimbrites. The average total magnetisation of normally magnetised rocks in the TVZ is about 1.7 A/m with 20° declination and –64° inclination (Soengkono 1990). The mean magnetisation of the upper, normally magnetised layer in the study area lies therefore between 1.7 and 2.6 A/m; for interpretation we used a weighted mean of 2 A/m. The vector average of total magnetisation for the reversely magnetised middle layer is about 0.4 A/m with 145° declination and +74° inclination (excluding the effect of local Titiraupenga Andesite).

Since the mean total magnetisation of the middle group of the older reversely magnetised rocks is small (i.e. 0.4 A/m), the contact between reversely magnetised rocks and nonmagnetic greywacke basement (i.e. the base of the reversely magnetised rocks) could not be modelled using the aeromagnetic data only. The approximate structure of the greywacke subsurface, however, is known from interpretation of residual gravity anomalies (Rogan 1982; Stern 1986; Soengkono 1990). In addition, some independent depth control is available from a seismic refraction survey across the Mangakino VC (Stern 1986) which indicates a minimum basement level of 2 km (below sea level). The residual gravity anomalies across the Mangakino VC along our magnetic profile LN19 are shown in the upper part of Fig. 3, together with the residual magnetic anomalies. The total thickness of

volcanic rocks along the section based on a 3-D gravity modelling using the algorithm of Barnett (1976) are shown in the lower part of Fig. 3. For the gravity model, a weighted mean density contrast of -0.5×10^3 kg/m^3 between volcanic rocks and the greywacke basement was assumed. Any decrease in density of the volcanic rocks in the lower part of the section was neglected. Our density contrast value of -0.5×10^3 kg/m^3 is similar to the weighted mean of -0.485×10^3 kg/m^3 indicated by a depth variable density section controlled by a seismic survey (Stern 1986) along an east–west trending profile running through Tihoi where basement also lies at a level of about 1–2 km below sea level. It is likely that the basement depths shown in Fig. 3 are minimum depths; the computed magnetic anomalies are not affected by this uncertainty.

Second-order residual magnetic anomalies were obtained by computing the magnetic effect of the volcanic infill, assuming a homogeneous reversed magnetisation of 0.4 A/m and by subtracting this effect from the observed residual. The second-order residuals were interpreted in terms of a normally magnetised surface layer (2.0 A/m). For both computations, the 3-D magnetic modelling algorithm of Barnett (1976) was used. The resulting best fit magnetic model across LN19 is shown in the lower part of Fig. 3; the resulting magnetic anomalies of both bodies are shown in the upper part. The relative error in the level of the interface separating normally and reversely magnetised rocks is large, probably of the order of 30%, as indicated by different models providing acceptable fits.

The same analysis of magnetic and gravity data along other east–west trending profiles across the area in Fig. 2 produced sections similar in structure to that shown in Fig. 3. We found no evidence that normally magnetised rocks are thicker than about 0.6 km in the Mangakino area. We also found that the widespread negative anomalies shown in Fig. 1 are associated either with contacts of nonmagnetic basement and older (>0.73 Ma) reversely magnetised rocks or with contacts of normally and reversely magnetised rocks. Although we could not define the deeper level of the bottom part of the reversely magnetised rocks by magnetic modelling, a wavelength analysis showed that these older rocks attain a thickness >1 km near moderately dipping (10–15° to the east) basement contacts at the western edge of the TVZ. Such thickness is required to explain the large wavelength magnetic anomaly between east–west co-ordinates 2735 and 2740 in the upper part of Fig. 3. Basement contact anomalies of similar wavelength also occur to the east of Titiraupenga and to the west of Tihoi (see Fig. 2).

Our magnetic stratigraphy model of the western TVZ (Table 3) shows that, with the exception of the Rocky Hill Ignimbrite, all deeper magnetic rocks are reversely magnetised and that the deepest (oldest) ignimbrite unit (Ignimbrite F) is in direct contact with the nonmagnetic basement in one part of the study area. It can be inferred that all volcanic rocks lying above the basement and below the normally magnetised surface layer are between 0.73 and 1.6 Ma old and constitute the bulk of all volcanic rocks in the western TVZ as shown in Fig. 2.

The structure of the upper layer of the magnetic model in Fig. 3 defines approximately the lateral extent of the Mangakino VC where younger (<0.73 Ma) pyroclastic flows can attain a thickness of the order of 0.5 km. This structure probably reflects subsidence (Soengkono 1990), presumably caused by secular compaction of older, nonwelded pyroclastics at depths >0.5 km. Subsidence, however, was not uniform, as indicated by a highstanding ridge (i.e. sample point PL23) inside the inferred volcanic centre (see Fig. 3). The rather smooth subsurface of the reversely magnetised rocks beneath most of the Mangakino VC explains the absence of any large negative magnetic anomalies. The small magnetic signature of this volcanic centre is therefore not caused by nonmagnetic deeper acidic magma as proposed by Rogan (1982); nor does it reflect entirely compensation of the effects of normally and reversely magnetised volcanic deposits as suggested by Wilson et al. (1984); the small magnetic signature is most likely the result of rather smooth, subparallel magnetic interfaces.

SUMMARY

Our paleomagnetic results, together with the new K-Ar dates, provide additional data for the stratigraphy of ignimbrites in the western TVZ between Tihoi and Mangakino. An approximate gross stratigraphy of these ignimbrites has now been established (Table 3). The eight older and four younger pyroclastic units classified by Wilson (1986a, b) were all erupted during the Matuyama geomagnetic epoch and are therefore older than 0.73 Ma; three of the older units could be dated by the K-Ar method, and their radiometric ages (1.06–1.60 Ma) confirm the gross magnetic stratigraphy.

Our initial hypothesis that reversely magnetised rocks are widespread in the study area has been confirmed by our laboratory studies of magnetic parameters and by magnetic interpretation models. The extent of reversely magnetised rocks in the western TVZ is therefore greater than that indicated by earlier studies (e.g., Cox 1971). An integrated interpretation of aeromagnetic and gravity data suggests that the maximum total thickness of reversely magnetised rocks in the study area is >1.5 km. Most of the volcanic rocks deposited in the western TVZ are made up by a thick sequence of older, reversely magnetised pyroclastic and lava flows. Younger, normally magnetised rocks occur in the form of a ≤0.6 km thick cover which outlines an older subsidence structure over the Mangakino VC.

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